

Anthony J. BarreraEmail: ajbarr116@gmail.com

Phone: 201-232-6304

LinkedIn: [Anthony Barrera - https://www.linkedin.com/in/anthony-barrera-93a361190/](https://www.linkedin.com/in/anthony-barrera-93a361190/)GitHub: [AnthonyBarrera116 - https://github.com/AnthonyBarrera116](https://github.com/AnthonyBarrera116)Website: <https://anthonybarrera116.github.io/Ajbarr.github.io/>

Education**Loyola University of Maryland**

B.S Computer Science

Minor: Statistic

Baltimore, MD

September 2019 - May 2023

September 2019 - May 2023

New Jersey Institute of Technology

M.S Computer Science

Newark, NJ

September 2023 – December 2024

Professional Experience**New Jersey Institute of Technology***User Services Support Specialist*

Newark, NJ

June 2026 – Present

- Manage inbound and outbound gate operations for truck drivers picking up and unloading cargo at the port.
- Monitor camera systems and gate activity to prevent truck backups, improve traffic flow, and support safe operations.
- Developed database-driven report automations to speed up reporting processes and reduce manual work.
- Collaborate with IT teams on technology improvements, process enhancements, and system upgrades to improve departmental efficiency.

New Jersey Institute of Technology*User Services Support Specialist*

Newark, NJ

October 2023 – December 2023

- Resolved an average of 25+ support tickets weekly for a population of over 8,000 users.
- Configured and maintained 300+ hardware/software systems across classrooms and labs.
- Streamlined technical documentation and improved turnaround time for common IT requests by 30%.

Loyola University Maryland*Information Technology Support Specialist*

Baltimore, MD

September 2019 – May 2023

- Diagnosed and resolved issues for 200+ unique users in a fast-paced university setting.
- Set up and deployed over 120 faculty and student workstations, reducing average setup time by 20%.
- Contributed to restructuring service response workflows, improving ticket resolution efficiency by 15%.

USA HOCKEY*Hockey Referee*

Various Rinks in NJ

September 2016 – September 2022

- Officiated 400+ youth games for players aged 5–18 with zero reported safety violations
- Coordinated with over 30 coaches and 100+ parents to maintain game integrity and enforce USA Hockey regulations.

Skills**Programming Languages (Advanced):** Python, C, C++, JavaScript**Programming Languages (Intermediate):** Java, SQL, Bash**Programming Languages (Basic):** Ruby, Solidity, HTML/CSS**Related Technology:****App:** React Native**Website:** HTML, React.js, Node.js, Express**Web 3:** Solidity, Hardhat, React.js**Database:** MongoDB, Oracle, MySQL**AI/ML:** PyTorch, TensorFlow, Torch, Keras, Scikit-learn, Gym, Gymnasium**Tools:** Git, GitHub Actions, VS Code, Android Studio, SQL Developer, XAMPP, Figma, Arduino IDE**OS:** Windows, MacOS, Linux**Concepts:** OOP, RESTful APIs, Agile, Unit Testing, Version Control

Projects

(Personal Project)

Personal Time

[Port Operations Reporting & Analytics Application](#) (Python, Pandas, Matplotlib, React Native Web, PyInstaller), 2026

2026 - Present

- Independently designed and coded the application outside scheduled work hours as a personal project.
- Demonstrated the application to operations staff as a proof of concept for simplifying recurring reports and reducing manual spreadsheet work.
- Built data import, analysis, charting, and export workflows for gate activity, lane performance, clerk productivity, daily empty, and recap reporting.
- Packaged the application as a Windows desktop program with local CSV storage, backups, and organized report exports.

(Personal Project)

Personal Time

[Stock Market Risk and Data analysis](#) (Python with ML/DL Libraries), ongoing

December 2024 - July 2025

- Gathering of data from the web microtrends using webdriver to scrape.
- Downloads and obtains day by day stock.
- To make an AI model to help predict in time to buy, sell or wait for stock using data scraped.

(Natural Language Processing)

New Jersey Institute of Technology

[BERT, Logistic Regression, SVC and etc using Sklearn \(Scratch\)](#) (Python with ML/DL Libraries), 2024 NJIT

September 2024 - December 2024

- Built and evaluated BERT, Logistic Regression, and SVC models for text classification using financial datasets.

(Deep Learning)

New Jersey Institute of Technology

[Image Animal classification CNN model \(Scratch\)](#) (Python with ML/DL Libraries), 2024 NJIT

September 2024 - December 2024

- Using ML/DL libraries to make CNN from scratch on image classification of 100 animals.
- Achieved 75% Test accuracy through GPU-accelerated training on +40,000 images.

(Data Mining)

New Jersey Institute of Technology

[K means and Hierarchical Clustering](#) (Python with basic numpy/pandas Libraries), 2024 NJIT

May 2024 - August 2024

- Basic K means and Hierarchical coded from scratch.
- Generated range to make points, and clustering and classifying to a # of groups. (accuracy 90%)

(Data Mining)

New Jersey Institute of Technology

[Apriori Algorithm, Brute Force, Association Rules](#) (Python with basic numpy/pandas Libraries), 2024 NJIT

May 2024 - August 2024

- Apriori, Brute Force, and Association Rules coded from scratch.
- Generated grocery list of 100 different combinations to find support, confidence and association rules to transactions.

(Artificial Intelligence)

New Jersey Institute of Technology

[AI-Stock Prediction Models](#) (Python with ML/DL Libraries), 2024 NJIT

January 2024 - May 2024

- Made 8 different models for stock prediction on 5 stocks. (Nvidia, Amazon, Tesla, AMD and Hosts Hotels and Resort Inc)
- 2 used a basic sklearn library with min/max and standard scaler. (accuracy: 50% more tests > lower %)
- 4 used min/max and standard on one model prediction close. (accuracy: 20-30% more tests > lower %)
- 2 used min/max and standard using 3 models, predicting high and low then close price. (accuracy: 20% more test > lower %)

(Data Management Systems Design)

New Jersey Institute of Technology

[Zoo Database Project](#) (HTML, Python XAMPP), 2023 NJIT

September 2023 - December 2023

- Designed a relational database tracking employee roles, animal housing, and facility assignments.
- Used XAMPP to connect to a DB and store our relationships.

(Senior Design Project)

Loyola University Maryland

[Finance Calculator Website](#) (HTML, JavaScript, MongoDB), 2023 Loyola

January 2023 - May 2023

- Designed a financial calculator used by 25+ finance students for TVM and capital budgeting problems.
 - Stored over 500 calculation history entries using JavaScript with MongoDB backend.
-

Projects (Continued)

(Cryptocurrencies and Blockchain)

Loyola University Maryland

[Crypto Token & Faucet Web App](#) (Solidity, JavaScript), 2023 Loyola

January 2023 - May 2023

- Launched an ERC-20 token and faucet on Sepolia Testnet used by 25+ student testers. (Faucet)
- Wrote and deployed 3 smart contracts integrated with a React.js frontend and Node.js backend.
- Handled wallet integration and token faucet deployment using MetaMask.

(Software Engineering)

Loyola University Maryland

[Recreational Football App](#) (JavaScript, React, Node.js, MongoDB), 2022 Loyola

September 2022 - December 2022

- Led development of a mobile app supporting 5+ game modes, 10+ team management tools.
- Increased beta user engagement by 40% after UI redesign using React Native and MongoDB.

Courses

• CS 496: Senior Project	Loyola University Maryland
• CS 482: Software Engineering	Loyola University Maryland
• CS 484: Machine Learning	Loyola University Maryland
• CS 489.01: Intro to Quantum Computing	Loyola University Maryland
• CS 489.02: Cryptos and Blockchain	Loyola University Maryland
• CS 403: Discovering Info in Data	Loyola University Maryland
• CS 479: Intro Microprocess-Based Systems	Loyola University Maryland
• CS 630: Operating Systems Design	New Jersey Institute Of Technology
• CS 656: Internet and Higher Layer Protocols	New Jersey Institute Of Technology
• CS 631: Data Management Systems Design	New Jersey Institute Of Technology
• DS 669: Reinforcement Learning	New Jersey Institute Of Technology
• CS 670: Artificial Intelligence	New Jersey Institute Of Technology
• CS 610: Data Structures and Algorithms	New Jersey Institute Of Technology
• DS 634: Data Mining	New Jersey Institute Of Technology
• DS 675: Machine Learning	New Jersey Institute Of Technology
• DS 677: Deep Learning	New Jersey Institute Of Technology
• DS 680: Natural Learning Processing	New Jersey Institute Of Technology
